Firat Adem Bilge

AI Engineer

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PROFILE

AI Engineer with **3+** years of experience in **machine learning**, distributed systems**,** and backend development. Developed an **AI agent** with RAG, a **distributed system** processing 600+ TB of data, and a **fingerprint segmentation model** reaching 86% IoU. **Published** in Springer Nature SIViP (2024). Experienced integrating **LLMs** and **RAG** into **backend services**.

EXPERIENCE

AI Engineer **Dec 2025 – Aug 2026**

Entervia — Part -Time *Milan, Italy*

- AI Support Agent — Built a Telegram support agent with **LangChain** and OpenAI API, resolving **50%** of support queries.
- RAG Pipeline — Built hybrid **pgvector** retrieval with reranking over sessions and FAQs, reaching **96%** retrieval accuracy.
- Tool Calling — Integrated **FastAPI** APIs as agent tools to check bookings, verify payments and send secure invitations.
- Agent Evaluation — Tested RAG accuracy, tool calls and failures on **87** cases with **LangSmith** and pytest, **94%** pass rate.
- Cloud Deployment — Launched agent on **AWS ECS** and **RDS** with LangSmith tracing and CI/CD, keeping **99.9%** uptime.

Software Engineer **Jun 2023 – Mar 2025**

Havelsan — Full-Time *Ankara, Turkey*

- Distributed Computing — Built a **Dask** platform on Linux to run video analytics on **600+** TB data, scaling **400+** workers.
- Kubernetes Deployment — Packaged Docker services as Helm charts and deployed to a **300+** node Kubernetes cluster.
- Backend Services — Developed **5** Python gRPC microservices with **PostgreSQL** metadata storage for video analytics.
- Event Streaming — Implemented **Kafka** queues and partitions for parallel video analytics, handling **300K+** requests/day.
- Observability — Deployed Prometheus, Grafana and **Elasticsearch** on nodes and built monitoring and diagnostic tools.

Machine Learning Engineer **Jun 2022 – Jun 2023**

Havelsan — Full-Time *Ankara, Turkey*

- Computer Vision — Developed a **PyTorch** segmentation model on **10K+** clean and latent fingerprints, reaching **86%** IoU.
- Model Selection — Compared 5 segmentation architectures by IoU on fingerprint images; U-Net scored best at **82%**.
- Image Preprocessing — Applied **OpenCV** morphology to isolate regions and curated hard latent prints to improve IoU.
- Model Optimization — Fine-tuned U-Net with Dice loss, pretrained encoder and augmentation, raising IoU by **4** points.

PROJECTS

PracticaMIA — Italian Bureaucracy AI Agent

- Agentic AI — Developed a **LangChain** agent with Python and FastAPI to automate key Italian administrative workflows.
- RAG Pipeline — Integrated **PostgreSQL** and **pgvector** for hybrid search of official requirements with source citations.

EDUCATION

Politecnico di Milano – *MSc in Computer Science and Engineering* **Apr 2025 – Apr 2027**

Hacettepe University – *BSc in Computer Science* **Oct 2017 – May 2022**

Czestochowa University of Technology – *Erasmus+ Exchange in Computer Science* **Sep 2019 – Feb 2020**

PUBLICATIONS

Tinysign: Sign Language Recognition in Low Resolution Settings **Jun 2024**

Signal, Image and Video Processing (SIViP), Springer Nature, DOI *London, United Kingdom*

- Developed a pose-conditioned super-resolution module and I3D classifier in **PyTorch**, reaching **47.5%** test accuracy.

CERTIFICATIONS

- **AI Engineer for Developers Associate — DataCamp** **Nov 2026**

SKILLS

Programming, Backend & Databases: Python, FastAPI, gRPC, Microservices, PostgreSQL

Generative & Agentic AI: LLMs, RAG, Tool Calling, LangChain, LangSmith, LLM Evaluation, MCP, pgvector, OpenAI API

Machine Learning & Data Engineering: PyTorch, OpenCV, Pandas, Dask, Apache Kafka

Cloud & DevOps: AWS (S3, Lambda, ECS, RDS), Docker, Kubernetes, CI/CD, Jenkins, Git, Linux, pytest

Spoken Languages: Turkish (Native), English (C1), Italian (B1)